Star Software

THE ENTERPRISE GUIDE TO INTELLIGENT DOCUMENT PROCESSING

From documents to trusted, validated, decision-ready data

The real automation challenge isn't extracting data from documents.
It's getting trusted, validated data into the system where decisions happen.

MTRs • COAs • Invoices • Insurance Forms • Transcripts • Shipping Documents • Supplier Certificates

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Executive Summary

Documents remain a critical input to enterprise operations even as core systems become increasingly digital. Invoices, certificates, claims forms, transcripts, shipping records and supplier documents often arrive as PDFs, scans, images, spreadsheets or semi-structured files. The information inside them may be essential to payment, quality, compliance, customer service, production and supply-chain decisions.

The first generation of document automation focused on OCR: turning pixels into text. That was useful, but it did not solve the business problem. A system can read a heat number correctly and still fail if it cannot determine whether the number belongs to the right material, whether the certificate meets the required specification, whether the value conflicts with another field, or where the validated result should go.

Intelligent Document Processing (IDP) closes that gap. A production-grade IDP workflow combines document ingestion, OCR, classification, extraction, validation, business rules, human review, workflow orchestration and downstream integration. The goal is not simply a better digital copy of a document. The goal is a trusted business record that can safely drive the next action.

This guide presents a practical framework for designing enterprise IDP across multiple document classes—from MTRs and COAs to invoices, insurance forms, transcripts, shipping documents and supplier certificates.

Why Document Automation Projects Stall

Many document projects begin with an attractive promise: upload a PDF, extract fields and eliminate manual entry. The proof of concept may perform well on a clean sample. Production is different. Supplier layouts change. Pages are missing. Scans are poor. Tables are dense. A field that looks correct in isolation fails a business rule when compared with another system.

The hidden complexity sits between reading and acting. Enterprise automation must answer five questions:

- What document is this?
- What information matters for this workflow?
- Is the extracted information correct and business-valid?
- What should happen when confidence is low or rules fail?
- Where should trusted data go, and what decision or action should it trigger?

That is why IDP should be designed as an end-to-end operating capability—not an OCR feature.

The IDP Pipeline: From Document to Decision

DOCUMENT → OCR → CLASSIFICATION → EXTRACTION → VALIDATION → BUSINESS RULES → HUMAN REVIEW → ERP / QMS / LIMS

Each stage has a distinct responsibility. Skipping a stage or treating it as optional can move errors downstream rather than remove them.

Stage	Primary question	Output
Ingestion / OCR	Can the system reliably read the source?	Text, layout, tables, images, metadata
Classification	What kind of document is this?	Document type, source, routing context
Extraction	Which fields, tables and entities matter?	Structured candidate data + confidence
Validation	Does the extracted data make sense?	Validated / failed fields and evidence
Business rules	Does it meet the organization's requirements?	Pass, exception, escalation or action
Human review	Which cases require judgment?	Corrected/approved data + reviewer trail
Integration	Where should trusted data go?	ERP/QMS/LIMS/workflow record

Layer 1 — Ingestion & OCR

Enterprise inputs are rarely uniform. A robust intake layer should handle native PDFs, scanned PDFs, images, spreadsheets and multi-page document packages. OCR converts visual information into machine-readable content while preserving page, position and structural context wherever possible. This matters because downstream validation may need to show exactly where a value came from.

Key design questions: Can the system handle poor scans? Does it preserve tables and page context? Can it process multi-page certificates or mixed document packets? Can it retain the original document and its metadata for audit purposes?

Layer 2 — Classification

Classification determines which extraction logic, schema and business rules should apply. A mixed inbound packet might contain an invoice, packing list, bill of lading and certificate. Treating every page as the same document creates avoidable errors.

Classification can use layout, text, visual features, metadata and contextual signals. Mature systems also support confidence-based routing and document splitting so that uncertain or mixed cases can be reviewed rather than silently misclassified.

Layer 3 — Extraction

Extraction converts document content into structured data. The right question is not 'How many characters did we read correctly?' but 'Did we capture the fields and tables the business process actually needs?'

For an MTR, that may include heat number, material grade, dimensions, chemical composition, mechanical properties and certification details. For an invoice, it may include supplier, invoice number, PO, line items, quantities, tax and totals. For a transcript, it may include student identity, courses, credits, grades and dates.

Layer 4 — Validation & Business Rules

Validation is where document data becomes business data. Extraction confidence alone is not enough: a model can be highly confident about a value that is nevertheless wrong for the transaction.

Validation should operate at multiple levels: field format checks; required-field checks; cross-field consistency; reference-data checks; specification checks; duplicate detection; and rules that compare document data with ERP/QMS/LIMS records. For quality certificates, this can mean checking material grade, heat number and measured properties against applicable requirements. For invoices, it can mean matching supplier, PO, quantities and totals.

Layer 5 — Human Review & Exception Management

Human-in-the-loop is not a failure of automation. It is a control mechanism. The objective is to let automation handle predictable cases while directing uncertain, high-risk or rule-failed cases to the right reviewer.

A well-designed review queue should show the extracted value alongside its source evidence, identify why the case was routed, capture the correction or approval, and preserve the decision in the audit trail. Human review should be selective—not a second manual data-entry process.

Layer 6 — Integration with ERP/QMS/LIMS

The automation outcome is incomplete if validated data stops in a dashboard or spreadsheet. The final stage is delivering trusted information into the system where work continues.

Depending on the workflow, that destination may be an ERP, QMS, LIMS, CRM, AP platform, claims system, student information system or workflow engine. Integration can use APIs, files, queues or other enterprise interfaces. The design objective is the same: remove unnecessary re-keying and make validated data available at the point of decision.

Applying IDP Across Enterprise Documents

Document type	Critical data	Validation / rules	Downstream action
MTRs	Heat/lot, grade, chemistry, mechanical properties	Specification, customer requirements, cross-field consistency	Quality record / ERP / traceability
COAs	Product identity, batch/lot, test results, certification	Specification and release criteria	QMS / customer documentation
Invoices	Vendor, PO, line items, tax, totals	PO match, duplicate check, tolerances	ERP / AP workflow / payment
Insurance forms	Policy, claimant, incident, coverage, supporting data	Completeness, policy rules, consistency	Claims workflow / review
Transcripts	Student, courses, grades, credits, dates	Identity, credit/grade rules, completeness	Student information / admissions
Shipping documents	Shipment, carrier, PO, quantities, dates	Order/quantity/reference checks	ERP / logistics / receiving
Supplier certificates	Supplier, material, lot, specifications, certifications	Approved supplier, specification, validity	QMS / ERP / procurement

A Reference Enterprise IDP Architecture

A scalable architecture separates document intelligence from business-system execution. This allows the same core capabilities—ingestion, classification, extraction, validation and review—to support multiple document workflows while keeping business rules and integration specific to each use case.

1. 1. Intake layer — email, portal, API, shared folder, scanner or application upload
2. 2. Document intelligence layer — OCR, layout understanding, classification, splitting and extraction
3. 3. Validation layer — confidence thresholds, deterministic checks, reference data and business rules
4. 4. Exception layer — human review, approvals, corrections, escalations and feedback
5. 5. Orchestration layer — workflow state, routing, notifications and retry handling
6. 6. Enterprise integration layer — ERP, QMS, LIMS, CRM, AP, claims, logistics or student systems
7. 7. Governance layer — access control, audit trail, retention, monitoring and model/process oversight

Governance, Security & Auditability

Enterprise IDP should be governed like any other system that creates or changes business records. Key controls include role-based access, protection of sensitive documents, retention policies, audit trails, environment separation, monitoring, exception reporting and clear ownership of business rules.

Provenance is particularly important. A reviewer or auditor should be able to trace a critical field from the downstream record back to the source document and, where practical, the location of the source value. This creates defensibility that a bare extracted data file cannot provide.

Measuring Business Value

A strong IDP business case measures the workflow, not just model accuracy. Baseline the current process before automation and compare the same measures after deployment.

Metric	What to measure	Why it matters
Cycle time	Document receipt to validated system record	Measures speed of business processing
Touch rate	% of documents requiring human intervention	Shows the real level of automation
Exception rate	% routed for review and primary causes	Reveals process/model improvement opportunities
Rework / correction rate	Corrections after initial processing	Measures data quality
Straight-through processing	% completed without manual intervention	Connects IDP to operational efficiency
Cost per document	Labor + technology + exception handling	Supports ROI
Decision latency	Time from document arrival to business action	Connects automation to business outcomes

Build vs. Buy

The right decision depends on document complexity, internal engineering capacity, integration requirements, governance needs and the number of workflows to be supported. A useful evaluation should go beyond OCR accuracy and ask whether the solution can handle real supplier/document variability, configurable validation, human review, auditability and production integration.

- Buy when speed to production, proven connectors, managed capabilities and reusable document intelligence are priorities.
- Build when the workflow is highly differentiated, internal platform capabilities are strong and long-term ownership is strategically important.
- Hybrid approaches can make sense when the enterprise wants to own orchestration and business rules while using specialized document intelligence components.

Enterprise Implementation Roadmap

Phase	Focus	Exit criteria
1. Discover	Map one high-value workflow; collect representative documents; baseline metrics	Defined scope, sample set, target fields, rules and KPIs
2. Pilot	Build the complete pipeline, including validation and review—not extraction alone	Measured accuracy, touch rate, exceptions and cycle time
3. Integrate	Connect validated outputs to the system of record	Reliable downstream transaction/record creation
4. Scale	Add suppliers, document types, rules and workflows	Reusable architecture, governance and monitoring
5. Optimize	Use exceptions and corrections to improve the process	Lower touch rate, faster cycle time and stronger controls

Enterprise Buyer's Checklist

☐ Handles real-world document variability—not only a fixed template set

☐ Supports OCR, classification, extraction and table/line-item processing

☐ Provides field-level confidence and source evidence

☐ Supports configurable validation and business rules

☐ Cross-checks document data against enterprise/reference data

☐ Routes exceptions to human reviewers with reason and evidence

☐ Preserves audit trail and document provenance

☐ Integrates with ERP, QMS, LIMS and other systems

☐ Supports security, access control and data-retention requirements

☐ Measures operational outcomes such as touch rate, cycle time and exception rate

☐ Supports new suppliers, formats and workflows without rebuilding the platform

☐ Has a practical pilot methodology with measurable acceptance criteria

Conclusion: The Document Is Not the Destination

The enterprise document problem is often described as a data-entry problem. It is bigger than that. Documents contain evidence that feeds operational decisions: whether material meets specification, whether an invoice should be paid, whether a claim should progress, whether a shipment can be received, or whether a student record is complete.

That is why the most useful IDP architecture is not 'OCR plus AI.' It is a controlled path from unstructured input to trusted business data: document → OCR → classification → extraction → validation → business rules → human review → ERP/QMS/LIMS.

Organizations that design around this complete pipeline can move beyond digitizing documents. They can create a dependable intelligence layer between the documents entering the enterprise and the systems where decisions happen.